作业

作业来自

梁昆淼《力学》 第 4 版 高等教育出版社

Johns Analytical Mechanics 2nd ed Oxford University Press

例如： L01.08 表示梁昆淼书第 01 章第 08 题； J02.05 Johns 书第 02 章第 05 题

J03.07† 表示 Johns 书第 03 章第 07 题，带 † 表示仅要求荣誉班同学完成

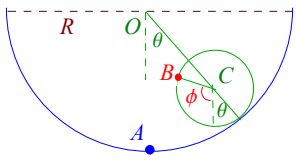
Homework 04, Week 06 2023.10.09 Due 2023.10.16 L03.03, L03.05, L03.06, L03.11, L03.14, L03.15

★ L03.06

半径为 r 的匀质柱在半径为 R 的柱形槽内作无滑动滚动，写出这种滚动的拉格朗日方程，

试求这匀质柱在槽底附近作小幅度摆动的周期

Solution



设 B 点与 A 点在小柱在最低点时重合

θ, ϕ 定义如图

θ 为 OC 与竖直线之夹角，

ϕ 为 CB 与竖直线之夹角，

无滑动滚动，故有

$$(\phi+\theta)\,r=\theta\,R \qquad (\dagger)$$

对小柱写出拉格朗日：$L=\dfrac{m}{2}(R-r)^2\dot{\theta}^2+\dfrac{1}{2}\left(\dfrac{1}{2}m\,r^2\right)\dot{\phi}^2+m\,g\,(R-r)\cos\theta$

利用式 (†) 消去 ϕ，可得拉格朗日量 $L=\dfrac{3}{4}m\,(R-r)^2\dot{\theta}^2+m\,g(R-r)\cos\theta$

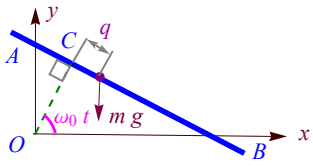
拉格朗日方程：$\dfrac{d}{dt}\dfrac{\partial L}{\partial\dot{\theta}}-\dfrac{\partial L}{\partial\theta}=0 \implies \dfrac{3}{2}m\,(R-r)^2\ddot{\theta}+m\,g(R-r)\sin\theta=0$

$$\theta\ll 1 \xrightarrow{\sin\theta\sim\theta} 3\,(R-r)\,\ddot{\theta}+2\,g\,\theta=0 \implies \omega^2=\frac{2\,g}{3\,(R-r)},\quad T=\pi\sqrt{6\,(R-r)/g}$$

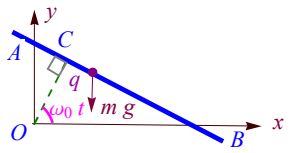
★ L03.11

细杆 AB 绕 O 点以角速度 ω_0 在竖直面里转动，求被束缚在杆上的一个质点的运动情况 $q\,(t)$

已知初始条件 $q(0)=0$, $\dot{q}(0)=0$



Solution



先建立杆坐标系与实验室坐标系之关系

设杆到坐标原点的距离 OC 为 l

质点距杆子上 C 点距离为 $q(t)$，如图

$$\begin{cases} x = l\cos\omega_0 t + q\sin\omega_0 t \\ y = l\sin\omega_0 t - q\cos\omega_0 t \end{cases}$$

质点的拉格朗日量

$$L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - m g y$$

$$\begin{cases} \dot{x} = \omega_0(-l\sin\omega_0 t + q\cos\omega_0 t) + \dot{q}\sin\omega_0 t \\ \dot{y} = \omega_0(l\cos\omega_0 t + q\sin\omega_0 t) - \dot{q}\sin\omega_0 t \end{cases}$$ 代入 L 得

$$L = \frac{1}{2}m\left[(\dot{q} - \omega_0 l)^2 + \omega_0{}^2 q^2\right] - m g\,(l\sin\omega_0 t - q\cos\omega_0 t)$$

拉格朗日方程：$\dfrac{d}{dt}\dfrac{\partial L}{\partial \dot{q}} - \dfrac{\partial L}{\partial q} = 0 \quad \Longrightarrow \quad \ddot{q} - \omega_0^2 q = g\cos\omega_0 t$

二阶线性常系数非齐次微分方程。先求齐次通解 $q_h = C_1 e^{\omega_0 t} + C_2 e^{-\omega_0 t}$

再求非齐次特解：$q_s = A\cos\omega_0 t \quad \Longrightarrow \quad A = -\dfrac{g}{2\omega_0^2}$

$$q = q_h + q_s = C_1 e^{\omega_0 t} + C_2 e^{-\omega_0 t} - \frac{g}{2\omega_0^2}\cos\omega_0 t$$

由 $q(0) = 0,\ \dot{q}(0) = 0 \quad \Longrightarrow \quad \begin{cases} C_1 + C_2 = g/(2\omega_0^2) \\ C_1 - C_2 = 0 \end{cases} \quad \Longrightarrow \quad q = \dfrac{g}{2\omega_0^2}(\cosh\omega_0 t - \cos\omega_0 t)$

★ L03.15

试证对于冲击作用，拉格朗日方程为

$$\left(\frac{\partial L}{\partial \dot{q}_a}\right)_{\text{end}} - \left(\frac{\partial L}{\partial \dot{q}_a}\right)_{\text{start}} = S_\alpha$$

其中 S_α 是广义冲击力的冲量，即：$S_\alpha = \displaystyle\int_{\text{start}}^{\text{end}} Q_\alpha^{(\text{impulse})}\,dt$

Solution

拉格朗日方程：$\dfrac{d}{dt}\dfrac{\partial L}{\partial \dot{q}_\alpha} - \dfrac{\partial L}{\partial q_\alpha} = Q_\alpha^{(\text{non})} + Q_\alpha^{(\text{impulse})}$

其中 $Q_\alpha^{(\text{non})}$ 为非保守力，$Q_\alpha^{(\text{impulse})}$ 为冲击力（也是非保守力）

两边同时积分 $\displaystyle\int_{\text{start}}^{\text{end}} dt$ 得

$$\left(\frac{\partial L}{\partial \dot{q}_\alpha}\right)_{\text{end}} - \left(\frac{\partial L}{\partial \dot{q}_\alpha}\right)_{\text{start}} = \int_{\text{start}}^{\text{end}} Q_\alpha^{(\text{non})}\,dt + \int_{\text{start}}^{\text{end}} Q_\alpha^{(\text{impulse})}\,dt + \int_{\text{start}}^{\text{end}}\left(\frac{\partial L}{\partial q_\alpha}\right)dt$$

方程右边第一和第三项大小有限，对冲击发生的很短时间的积分为 0

第二项冲击力很大，积分有限 $\displaystyle\int_{\text{start}}^{\text{end}} Q_\alpha^{(\text{impulse})}\,dt = S_\alpha$

故有：$\left(\dfrac{\partial L}{\partial \dot{q}_\alpha}\right)_{\text{end}} - \left(\dfrac{\partial L}{\partial \dot{q}_\alpha}\right)_{\text{start}} = \displaystyle\int_{\text{start}}^{\text{end}} Q_\alpha^{(\text{impulse})}\,dt = S_\alpha$

★ L03.03

匀质链，长度为 l，一半在光滑桌面，另一半垂挂与桌外，链无初速下滑，问

在链全部离开桌面的瞬间，其速度为多少

Solution

设桌上的一段链长度 x，链的拉格朗日量为

$$L=\frac{1}{2}m\dot{x}^2-\left[-\frac{(l-x)}{l}m\,g\,\frac{(l-x)}{2}\right]\quad\Longrightarrow\quad L=\frac{1}{2}m\dot{x}^2+\frac{1}{2}\frac{m\,g}{l}(l-x)^2$$

中括号内为桌外一段链的重力势能。

拉格朗日方程：$\frac{d}{dt}\frac{\partial L}{\partial\dot{x}}-\frac{\partial L}{\partial x}=0$

$$\Longrightarrow\quad\ddot{x}-\frac{g\,x}{l}+g=0\quad\Longrightarrow\quad x=l+C_1\,e^{t/\tau}+C_2\,e^{-t/\tau},\ \text{ with }\tau=\sqrt{l/g}$$

初条：$x(0)=l/2,\ \dot{x}(0)=0\quad\Longrightarrow\quad x=l\left(1-\frac{1}{2}\cosh\frac{t}{\tau}\right),\quad\dot{x}=\frac{l}{2\tau}\sinh\frac{t}{\tau}$

当 $x=0$ 时，$\cosh\frac{t_0}{\tau}=2,\quad\dot{x}=\frac{l}{2\tau}\sqrt{\cosh^2\frac{t_0}{\tau}-1}=\frac{\sqrt{3}}{2}\sqrt{l\,g}$

也可利用广义能量定理

$$H=\frac{\partial L}{\partial\dot{x}}\dot{x}-L=L=\frac{1}{2}m\dot{x}^2-\frac{1}{2}\frac{m\,g}{l}(l-x)^2$$

$$\frac{dH}{dt}=-\frac{\partial L}{\partial t}=0\quad\Longrightarrow\quad H(t)=H(0)=-\frac{1}{8}m\,g\,l\ \text{ with the use of }x(0)=l/2,\ \dot{x}(0)=0$$

在链全部离开桌面的瞬间，$x=0$

$$H(t)=H(0)\quad\Longrightarrow\quad\frac{1}{2}m\dot{x}^2-\frac{1}{2}m\,g\,l=-\frac{1}{8}m\,g\,l\quad\Longrightarrow\dot{x}=\frac{\sqrt{3}}{2}\sqrt{l\,g}$$

所以，广义能量定理并未能给出新方程，只是让求解过程简化。

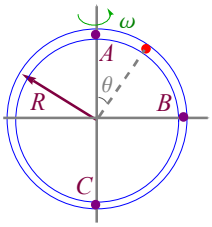
利用能量守恒求解的问题在拉格朗日力学中可以用广义能量定理求解。

```
xt = l + C1 Exp[t / τ] + C2 Exp[-t / τ];
eq1 = (xt /. t → 0) - l / 2;
eq2 = (D[xt, t] /. t → 0);
Solve[{eq1 == 0, eq2 == 0}, {C1, C2}]
```

$$\left\{\left\{C1\to-\frac{l}{4},\ C2\to-\frac{l}{4}\right\}\right\}$$

★ L03.05

空心圆环可绕竖直轴 AC 转动，转动惯量为 I，环半径 R，环的初始角速度为 ω_0，质量为 m 的质点静止于环内最高点 A 。由于微小的干扰，小球向下滑动，问质点滑到 B 点和 C 点时，环的角速度和质点速度各是多大？设环内壁光滑。

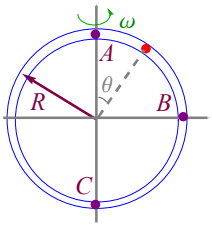


Solution

把圆环和质点看成一个体系，
取 θ 和圆环转过的角度 ϕ 作为广义坐标
拉格朗日量：

$$L = \frac{1}{2} I \dot{\phi}^2 + \frac{1}{2} m \left[\left(R \dot{\theta}\right)^2 + \left(R \sin\theta \dot{\phi}\right)^2 \right] - m g R \cos\theta$$

其中：第一项为圆环转动动能，环心为势能零点，
质点速度的两分量分别为 $R\dot{\theta}$ 和 $R \sin\theta \dot{\phi}$



这其实已经是一个约化的拉格朗日量（记得利用约化拉格朗日量无法求得约束力）。以下求拉格朗日方程

ϕ 是可遗坐标，故有：$\frac{d}{dt}\frac{\partial L}{\partial \dot{\phi}} = \frac{\partial L}{\partial \phi} = 0$

广义动量 $p_\phi = \frac{\partial L}{\partial \dot{\phi}}$ 是个守恒量，$p_\phi = \left(I + m R^2 \sin^2\theta\right)\dot{\phi} = I \omega_0$ (1)

接着，$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0 \implies R\left(\ddot{\theta} - \sin\theta\cos\theta\,\dot{\phi}^2\right) - g\sin\theta = 0$ (2)

以上对 2 个自由坐标给出2 个拉格朗日方程。

“人如月，对花笑，还有谁？” ——广义能量定理

$$H = \frac{\partial L}{\partial \dot{\phi}}\dot{\phi} + \frac{\partial L}{\partial \dot{\theta}}\dot{\theta} - L = \frac{1}{2}\left(I + m R^2 \sin^2\theta\right)\dot{\phi}^2 + \frac{1}{2} m R^2 \dot{\theta}^2 + m g R \cos\theta$$

$$\frac{dH}{dt} = -\frac{\partial L}{\partial t} = 0 \implies H = H(t = 0)$$

$$\implies \frac{1}{2}\left(I + m R^2 \sin^2\theta\right)\dot{\phi}^2 + \frac{1}{2} m R^2 \dot{\theta}^2 + m g R \cos\theta = \frac{1}{2} I \omega_0^2 + m g R \quad (3)$$

方程(1 – 3) 只有 2 个是独立的，选其中任意 2 个求解。通常选运动积分（守恒量）的方程。下选方程 (1) 和 (3) 求解。

利用 $\theta_B = \frac{\pi}{2} \xrightarrow{\text{方程 (1) 和 (3)}} \begin{cases} \left(I + m R^2\right)\omega_B = I \omega_0 \\ \left(I + m R^2\right)\omega_B^2 + m R^2 \dot{\theta}_B^2 = I \omega_0^2 + 2 m g R \end{cases}$

$$\implies \begin{cases} \omega_B = \dot{\phi}_B = \dfrac{I \omega_0}{\left(I + m R^2\right)} \\ v_B^2 = R^2 \omega_B^2 + R^2 \dot{\theta}_B^2 = \dfrac{I R^2 \omega_0^2 \left(2 I + m R^2\right)}{\left(I + m R^2\right)^2} + 2 R g \end{cases}$$

若求质点相对于环的速度，则 $v'_B = R \dot{\theta}_B = \left[\frac{I R^2 \omega_0^2}{\left(I + m R^2\right)} + 2 R g\right]^{1/2}$

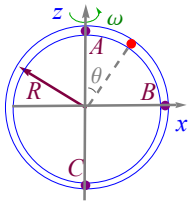
利用 $\theta_C = \pi \xrightarrow{\text{方程 (1) 和 (3)}} \begin{cases} I \omega_C = I \dot{\phi}_C = I \omega_0 \\ I \omega_C^2 + m R^2 \dot{\theta}_C^2 - 2 m g R = I \omega_0^2 + 2 m g R \end{cases}$

$$\implies \begin{cases} \omega_C = \omega_0 \\ v_C^2 = R^2 \dot{\theta}_C^2 = 4 R g \end{cases}$$

再次说明广义能量定理可以简化求解过程。

如果从方程 (2) 出发，还需利用 $\ddot{\theta} = \dot{\theta}\frac{d\dot{\theta}}{d\theta}$ 做个积分才能得到 $\dot{\theta}$ 与 θ 的关系

为什么到达 B 点时，质点竖直速度分量大于 $\sqrt{2 R g}$，什么力做的功影响了速度竖直分量？



注：圆环对质点的作用力有几个分量
一个是水平分量，是带着质点绕 z 轴 转动
使得质点到达 B 点时具有 y 向的速度
一个是竖直分量，在 AB 段是竖直向下。
这个分量的力和重力一起加速质点，
使得质点到达 B 点时的 z 向速度 $R\,\dot{\theta}_B$ 大于 $2\,R\,g$
尽管在圆环平面内的弹力总体而言不做功
但水平和竖直分量都做功，竖直分量增加 z 向速度
水平方向减小 x 向速度（否则质点就飞出去了）

那么，如果圆环不转动，质点到达 B 点的速度为何就是 $\sqrt{2\,R\,g}$ 了呢？

因为若圆环不转动，质点刚离开 A 点时弹力方向是远离圆环圆心的，
其竖直分量做负功，阻碍质点 z 向速度的增加

但其水平分量做正功，增加速度的水平分量，弹力改变竖直和水平速度的分配，尽管弹力作的总功为 0，

一正一负调整了水平和竖直方向的速度分量分配（但速度的大小不会变，由机械能守恒确定）

质点到达某个临界角时，弹力开始反向，变成指向圆环圆心（否则质点就飞出了），

弹力的竖直分量与水平分量分别作正功和负功，所以到达 B 点时，水平速度为 0，竖直速度为 $\sqrt{2\,R\,g}$

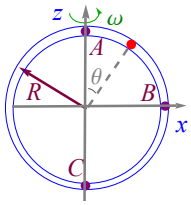
当圆环转动时，弹力必须提供质点圆周运动所需的向心力，从而弹力一开始就指向圆环圆心。

使得在 AB 段运动过程中，弹力总是和重力一起增大速度的竖直分量，从而到 B 点时，速度大于 $\sqrt{2\,R\,g}$

简而言之，圆环作用于质点的弹力在圆环平面内的分量尽管不做功，但其实它可以在分解为竖直分量和水平分量，

这两个分量在保证机械能守恒的前提下分别改变质点的竖直和水平速度。

约束力



约化的拉格朗日量：

$$\tilde{L} = \frac{1}{2} I\,\dot{\phi}^2 + \frac{1}{2} m \left[\left(R\,\dot{\theta}\right)^2 + \left(R \sin\theta\,\dot{\phi}\right)^2 \right] - m\,g\,R\cos\theta$$

$$\frac{d}{dt}\frac{\partial \tilde{L}}{\partial \dot{\phi}} = \frac{\partial \tilde{L}}{\partial \phi} = 0 \implies p_\phi = \frac{\partial L}{\partial \dot{\phi}} \text{ 是个守恒量，} \quad p_\phi = \left(I + m\,R^2 \sin^2\theta\right)\dot{\phi} = I\,\omega_0 \quad (\dagger)$$

$$\tilde{H} = \frac{\partial \tilde{L}}{\partial \dot{\phi}}\,\dot{\phi} + \frac{\partial \tilde{L}}{\partial \dot{\theta}}\,\dot{\theta} - \tilde{L} = \frac{1}{2}\left(I + m\,R^2 \sin^2\theta\right)\dot{\phi}^2 + \frac{1}{2} m\,R^2\,\dot{\theta}^2 + m\,g\,R\cos\theta$$

$$\frac{d\tilde{H}}{dt} = 0 \implies \frac{1}{2}\left(I + m\,R^2 \sin^2\theta\right)\dot{\phi}^2 + \frac{1}{2} m\,R^2\,\dot{\theta}^2 + m\,g\,R\cos\theta = \frac{1}{2} I\,\omega_0^2 + m\,g\,R \quad (\ddagger)$$

由方程 (†) 和 (‡) 求得：$\dot{\phi} = \dfrac{I\,\omega_0}{\left(I + m\,R^2 \sin^2\theta\right)} = \dfrac{\eta\,\omega_0}{\left(\eta + \sin^2\theta\right)}$,

$$R^2\,\dot{\theta}^2 = 2\,R\,g\,(1 - \cos\theta) + \frac{\eta\,R^2\,\omega_0^2 \sin^2\theta}{\left(\eta + \sin^2\theta\right)}, \qquad \eta = \frac{I}{m\,R^2}$$

选 s-系：$x,\ y,\ z$，　q-系：$r,\ \theta,\ \varphi$　（以球坐标描述质点位置）

变换：
$$\begin{cases} s_1 = x = r\sin\theta\cos\varphi \\ s_2 = y = r\sin\theta\sin\varphi \\ s_3 = z = r\cos\theta \\ s_4 = \phi \end{cases} \qquad \begin{cases} \dot{x} = \dot{r}\sin\theta\cos\varphi + r\,\dot{\theta}\cos\theta\cos\varphi - r\,\dot{\varphi}\sin\theta\sin\varphi \\ \dot{y} = \dot{r}\sin\theta\sin\varphi + r\,\dot{\theta}\cos\theta\sin\varphi + r\,\dot{\varphi}\sin\theta\cos\varphi \\ \dot{z} = \dot{r}\cos\theta - r\,\dot{\theta}\sin\theta \end{cases} \qquad \begin{cases} q_1 = r = \left(x^2 + y^2 + z^2\right)^{1/2} \\ q_2 = \theta = \arctan\left[\left(x^2 + y^2\right)^{1/2}/z\right] \\ q_3 = \varphi \\ q_4 = \phi \end{cases}$$

完整的拉格朗日量

$$s\text{-系：}\ L_s = \frac{I}{2}\,\dot{\phi}^2 + \frac{m}{2}\left(\dot{x}^2 + \dot{y}^2 + \dot{z}^2\right) - m\,g\,z$$

$$q\text{-系：}\ L_q = \frac{I}{2}\,\dot{\phi}^2 + \frac{m}{2}\left(\dot{r}^2 + r^2\,\dot{\theta}^2 + r^2\,\dot{\varphi}^2 \sin^2\theta\right) - m\,g\,r\cos\theta$$

```
xd = rd Sin[θ] Cos[φ] + r θd Cos[θ] Cos[φ] - r φd Sin[θ] Sin[φ];
yd = rd Sin[θ] Sin[φ] + r θd Cos[θ] Sin[φ] + r φd Sin[θ] Cos[φ];
zd = rd Cos[θ] - r θd Sin[θ];
Simplify[xd^2 + yd^2 + zd^2]
```

$$rd^2 + \frac{1}{2} r^2 \left(2\,\theta d^2 + \varphi d^2\right) - \frac{1}{2} r^2 \varphi d^2 \,\mathrm{Cos}[2\,\theta]$$

约束函数：$\begin{cases} G_1 = r - R \\ G_2 = \varphi - \phi \end{cases}$ $\quad L_q = \frac{I}{2}\dot{\phi}^2 + \frac{m}{2}\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\dot{\varphi}^2\sin^2\theta\right) - m\,g\,r\cos\theta,\quad q_1 = r,\ \ q_2 = \theta,\ \ q_3 = \varphi,\ \ q_4 = \phi$

q-系的拉格朗日方程：$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_\beta} - \frac{\partial L}{\partial q_\beta} = \lambda_1 \frac{\partial G_1}{\partial q_\beta} + \lambda_2 \frac{\partial G_2}{\partial q_\beta}$ 现有 2 个约束

$$\begin{cases} q_\beta = r\text{ 时：}\ m\ddot{r} - m\,r\dot{\theta}^2 - m\,r\dot{\varphi}^2\sin^2\theta + m\,g\cos\theta = \lambda_1 & \xRightarrow{\text{约束条件}} \quad \lambda_1 = m\,g\cos\theta - m\,R\dot{\theta}^2 - m\,R\dot{\varphi}^2\sin^2\theta \\ q_\beta = \theta\text{ 时：}\ m\frac{d}{dt}\left(r^2\dot{\theta}\right) - m\,r^2\dot{\varphi}^2\sin\theta\cos\theta - m\,g\,r\sin\theta = 0 & \xRightarrow{\text{约束条件}} \quad \ddot{\theta} = \dot{\varphi}^2\sin\theta\cos\theta + \frac{g}{R}\sin\theta \\ q_\beta = \varphi\text{ 时：}\ m\frac{d}{dt}\left(r^2\dot{\varphi}\sin^2\theta\right) = \lambda_2 & \xRightarrow{\text{约束条件}} \quad \frac{d}{dt}\left(m\,R^2\dot{\varphi}\sin^2\theta\right) = \lambda_2 \\ q_\beta = \phi\text{ 时：}\ \frac{d}{dt}\left(I\dot{\phi}\right) = -\lambda_2 & \xRightarrow{\text{约束条件}} \quad \frac{d}{dt}\left(I\dot{\varphi}\right) = -\lambda_2 \end{cases}$$

前已求得：$\dot{\varphi} = \frac{\eta\,\omega_0}{(\eta + \sin^2\theta)}$ 和 $R^2\dot{\theta}^2 = 2\,R\,g\,(1 - \cos\theta) + \frac{\eta\,R^2\,\omega_0^2\sin^2\theta}{(\eta + \sin^2\theta)}$

```
φd = w0/(1 + Sin[θ]^2);
θd = (2 g/R (1 - Cos[θ]) + w0^2 Sin[θ]^2/(1 + Sin[θ]^2))^(1/2);

t1 = D[θd, θ] θd;
t2 = φd^2 Sin[θ] Cos[θ] + g/R Sin[θ];

Simplify[t2 - t1]
```

```
0
```

$$\lambda_1 = m\,g(3\cos\theta - 2) - \frac{I\,\omega_0^2\left(2\,\eta + \sin^2\theta\right)\sin^2\theta}{R\left(\eta + \sin^2\theta\right)^2},\qquad \lambda_2 = -I\,\ddot{\varphi} = \frac{I\,\eta\,\omega_0\sin 2\,\theta}{\left(\eta + \sin^2\theta\right)^2}\dot{\theta}$$

记得，$\begin{cases} G_1 = r - R \\ G_2 = \varphi - \phi \end{cases}$，$\begin{cases} q_1 = r,\ \ q_2 = \theta,\ \ q_3 = \varphi,\ \ q_4 = \phi, \\ s_1 = x,\ \ s_2 = y,\ \ s_3 = z,\ \ s_4 = \phi \end{cases}$

由：$Q_\beta = \lambda_1 \frac{\partial G_1}{\partial q_\beta} + \lambda_2 \frac{\partial G_2}{\partial q_\beta} \quad \Longrightarrow \quad Q_1 = \lambda_1 \frac{\partial G_1}{\partial q_1} = \lambda_1,\quad Q_2 = 0,\quad Q_3 = \lambda_2,\quad Q_4 = -\lambda_2$

$$F_x = \sum_{\alpha=1}^{3} Q_\alpha \frac{\partial q_\alpha}{\partial s_1} = Q_1\frac{\partial r}{\partial x} + Q_2\frac{\partial \theta}{\partial x} + Q_3\frac{\partial \varphi}{\partial x} + Q_4\frac{\partial \phi}{\partial x} = \lambda_1\frac{\partial r}{\partial x} + \lambda_2\frac{\partial \varphi}{\partial x} = \lambda_1\sin\theta\cos\varphi - \lambda_2\frac{\sin\phi}{R\sin\theta}$$

$$F_y = \sum_{\alpha=1}^{3} Q_\alpha \frac{\partial q_\alpha}{\partial s_2} = Q_1\frac{\partial r}{\partial y} + Q_2\frac{\partial \theta}{\partial y} + Q_3\frac{\partial \varphi}{\partial y} + Q_4\frac{\partial \phi}{\partial y} = \lambda_1\frac{\partial r}{\partial y} + \lambda_2\frac{\partial \varphi}{\partial y} = \lambda_1\sin\theta\sin\varphi + \lambda_2\frac{\cos\phi}{R\sin\theta}$$

$$F_z = Q_1 \frac{\partial r}{\partial z} = \lambda_1 \cos\theta, \qquad F_4 = Q_4 = -\lambda_2$$

$\Longrightarrow$ 作用于质点的约束力 $\vec{F} = \lambda_1 \nabla r = \lambda_1 \hat{r} + \dfrac{\lambda_2}{R\sin\theta}\hat{\varphi}$, 作用于圆环的力矩 $L_{圆环} = -\lambda_2$

在 B 点：$\theta = \dfrac{\pi}{2}$, $\vec{F} = \lambda_1 \hat{r} = -\left[2mg + \dfrac{I\omega_0^2\,(2\eta+1)}{R(\eta+1)^2}\right]\hat{r} = -m\dfrac{v_B^2}{R}\hat{r}$

记得：$v_B^2 = \dfrac{I R^2 \omega_0^2 (2I + mR^2)}{(I + mR^2)^2} + 2Rg$,

在 C 点：$\theta = \pi$, $\vec{F} = \lambda_1 \hat{r} = 5mg\hat{z} = \left(mg + m\dfrac{v_C^2}{R}\right)\hat{z}$, 其中 $v_C^2 = 4Rg$

也可直接利用 $\vec{F} = \lambda_1 \nabla G_1 + \lambda_2 \nabla G_2$ 得到 $\vec{F} = \lambda_1 \vec{e}_r + \dfrac{\lambda_2}{R\sin\theta}\vec{e}_\varphi$

其中：$Q_\beta = \sum_{a=1}^{C} \lambda_a \dfrac{\partial G_a}{\partial q_\beta}$, $F_\beta = \sum_{\alpha=1}^{C} Q_\alpha \dfrac{\partial q_\alpha}{\partial s_\beta} = \sum_{a=1}^{C} \lambda_a \sum_{\alpha=1}^{C} \dfrac{\partial G_a}{\partial q_\alpha}\dfrac{\partial q_\alpha}{\partial s_\beta} = \sum_{a=1}^{C} \lambda_a \dfrac{\partial G_a}{\partial s_\beta}$

which yields $\vec{F} = \lambda_1 \nabla G_1 + \lambda_2 \nabla G_2$

★ L03.14

质量为 m 的质点连接于不可伸缩轻杆的一端，杆的另一端可绕固定点向任意方向自由运动，这一装置角球摆，求球摆的运动。

Solution

选取球坐标：$\begin{cases} x = r\sin\theta\cos\phi \\ y = r\sin\theta\sin\phi \\ z = r\cos\theta \end{cases}$ 拉格朗日量：$L = \dfrac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - mgz$

化到球坐标中：$\begin{cases} \dot{x} = \dot{r}\sin\theta\cos\phi + r\dot{\theta}\cos\theta\cos\phi - r\dot{\phi}\sin\theta\sin\phi \\ \dot{y} = \dot{r}\sin\theta\sin\phi + r\dot{\theta}\cos\theta\sin\phi + r\dot{\phi}\sin\theta\cos\phi \\ \dot{z} = \dot{r}\cos\theta - r\dot{\theta}\sin\theta \end{cases}$

$$\dot{x}^2 + \dot{y}^2 + \dot{z}^2 = \dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\phi}^2 \quad \Longrightarrow \quad L = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\phi}^2\right) - mgl\cos\theta$$

约束函数 $G_1 = r - l$ 代入拉格朗日量得约化拉格朗日量，$r = l$, $\dot{r} = 0$

$$\tilde{L} = \frac{1}{2}ml^2\left(\dot{\theta}^2 + \dot{\phi}^2\sin^2\theta\right) - mgl\cos\theta$$ （其实也可以直接写出这个约化的拉格朗日量）

$\Longrightarrow$ ϕ 是可遗坐标，$p_\phi = \dfrac{\partial\tilde{L}}{\partial\dot{\phi}} = ml^2\dot{\phi}\sin^2\theta = L_z$ 守恒 $\Longrightarrow$ $\dot{\phi} = \dfrac{L_z}{ml^2\sin^2\theta}$ (1)

关于 θ 的方程较为复杂，选择利用广义能量定理。

$$\Longrightarrow \quad H = \frac{\partial\tilde{L}}{\partial\dot{\phi}}\dot{\phi} + \frac{\partial\tilde{L}}{\partial\dot{\theta}}\dot{\theta} - \tilde{L} = \frac{1}{2}ml^2\left(\dot{\theta}^2 + \dot{\phi}^2\sin^2\theta\right) + mgl\cos\theta = E \text{ 守恒}$$

$$\Longrightarrow \quad \frac{1}{2}ml^2\dot{\theta}^2 + \frac{L_z^2}{2ml^2\sin^2\theta} + mgl\cos\theta = E$$

$$\Longrightarrow \quad \int_{t_0}^{t} dt = \pm\int_{\theta_0}^{\theta} \frac{d\theta}{\left[2\left(E - \dfrac{L_z^2}{2ml^2\sin^2\theta} - mgl\cos\theta\right)\Big/ml^2\right]^{1/2}}$$

利用初始条件，由上式求出 $\theta(t)$，再代入 (1) 式对时间 t 积分求得 $\phi(t)$。